INCH-POUND

A-A-59774

10 May 2005

COMMERCIAL ITEM DESCRIPTION

BREATHING AIR COMPRESSOR (BAC), PORTABLE, DIESEL

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE.

1.1 Scope. This commercial item description covers the portable, diesel breathing air compressor (BAC) used by Navy personnel to refill breathing air cylinders used with the open-circuit self contained breathing apparatus (SCBA). This equipment will be used aboard Navy ships and at Navy Training Centers. The portable, diesel BAC is not intended for use with oxygen bottling.

2. SALIENT CHARACTERISTICS.

2.1 Physical characteristics.

2.1.1 Size. The overall size of the BAC in it stowed configuration shall not exceed 48 inches long by 20 inches wide by 30 inches high.

2.1.2 Weight. The BAC shall have a dry weight of less than 250 lbs.

2.1.3 Frame. The BAC shall be configured in a tubular metal frame. The frame shall be configured to protect the BAC components from rough handling of the unit as well as provide for convenient lifting points when transporting the BAC. The frame shall be composed of a metal that provides resistance to corrosion in a sea/salt environment.

2.1.4 Carrying handles. The BAC frame shall be provided with four conveniently located carrying handles (each at diametrically opposed corners of the unit) to facilitate transporting the BAC aboard ship. The carrying handles may be retractable in order to maintain the specified size requirements for stowage of the BAC. With the handles extended, the overall size of the BAC shall not exceed 60 inches long by 20 inches wide by 30 inches high.

2.1.5 Casters. The BAC frame shall be mounted on four heavy-duty casters to facilitate transporting the unit on flat surfaces. Each caster shall have a nominal rated capacity of at least 250 lbs. The bracket and bearings must resist corrosion in a sea/salt spray environment.

2.1.6 Engine. An air-cooled diesel engine shall drive the BAC. The engine may be modified to provide some level of corrosion resistance when exposed to a corrosive environment.

2.1.7 Purification system. The BAC shall have a purification system that includes an oil-water separator, a condensate drain valve, and a breathing air purifier. A backpressure regulating valve and check valves shall be provided, as required, to ensure proper operation of the purification system. The breathing air purifier shall employ a single, user-replaceable air purifier cartridge with a processing capability of at least 15,000 standard ft^3 of air based on an inlet air temperature of 70 °F.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Commander, Naval Sea Systems Command, ATTN: SEA 05Q, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to commandstandards@navsea.navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil/.

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2.1.8 Fill hoses. The BAC shall be provided with two fill hoses (charging whips), which permit filling of either a single SCBA cylinder or two SCBA cylinders simultaneously. The end of each fill hose shall be provided with an air shut-off valve, a pressure gauge, an air bleed valve, a relief valve and a CGA-347 fill hose adapter to facilitate filling 4500 psig SCBA cylinders. Each fill hose shall be six feet in length (+/- six inches). The BAC shall also be equipped with corrosion resistant stowage brackets for the charging hoses.

2.1.9 Corrosion resistant high pressure fittings. The BAC shall be provided with corrosion resistant austenitic stainless steel tubing, tube fittings, high pressure instrument piping, and fill hose fittings. The CGA-347 fill hose adaptor fittings shall be either stainless steel or anodized aluminum.

2.1.10 Breathing inlet air filter and hose. A detachable breathing air inlet filter and inlet air hose shall be provided with the BAC. The nominal length of the inlet air hose shall be 10 feet (+/- one foot). The hose shall have some type of reinforcement incorporated into its design to prevent excess bending or kinking of the hose during stowage.

2.1.11 Hour meter. The BAC shall have an elapsed run time meter, incremented in hours. The meter shall operate when the diesel engine is running. The meter shall be able to operate properly when exposed to interference from the conditions that exist within a shipboard environment such as vibration from ship movements.

2.1.12 Moisture/CO level indicator. The BAC shall be equipped with a means of monitoring the moisture and CO levels.

2.1.13 Relief/safety valves. The BAC relief valves shall be tagged with the following information:

a. Pressure setting
b. Set date

2.1.14 Pressure gauges. The pressure gauges should be labeled with the calibration date.

2.2 Performance characteristics.

2.2.1 Air pressure. The BAC shall deliver air for charging breathing air cylinders to a nominal pressure of 4500 psig. The outlet air pressure setting shall be user-adjustable via a relief valve or similar device. The BAC shall be capable of delivering breathing air at a maximum pressure of 4950 psig.

2.2.2 Free air delivery. The BAC shall have a free air delivery of at least 5.0 standard cubic feet per minute (SCFM) at an outlet air pressure of 3000 psig.

2.2.3 Air quality. The BAC purification system shall be capable of delivering Grade D breathing air in accordance with CGA-G7.1 with a moisture content of 20 mg/m^3 or a dew point of -65 °F.

2.3 Design and construction.

2.3.1 Workmanship. The BAC and all accessories shall be free from damage such as cracks, burrs, sharp edges, tears, punctures, mars, abrasions, pits, and any other defects that will adversely affect the performance, reliability, safety, durability or appearance of the BAC. The BAC and all accessories shall be free from foreign material such as dirt, grease, oil, or metal chips. Rubber parts shall be compact (free from holes or bubbles) and free from foreign matter. Metal parts shall be free of distortion, burrs, sharp edges, scratches, or other damage. No voids shall be evidenced in any metal-to-rubber bond areas.

2.3.2 Interchangeability. Part and component interchangeability shall be inherent in the BAC design. All parts and assemblies assigned a unique part number shall be functionally and physically interchangeable between different BACs.

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2.4 Environmental requirements.

2.4.1 Shock. The BAC shall meet the shock requirements as specified in MIL-S-901, Grade B.

2.4.2 Vibration. The BAC shall meet the requirements for Type I environmental vibration as specified in MIL-STD-167-1. The exploratory variable range shall be 4 Hz to 30 Hz.

2.4.3 Operating temperature. The BAC shall be fully operational at an ambient temperature between 32 ˚F and 115 ˚F.

3. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. PRODUCT CONFORMANCE.

4.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Source of documents.

6.1.1 CGA. CGA specifications and standards are available from the Compressed Gas Association, Inc., 4221 Walney Road, 5th Floor, Chantilly, VA 20151-2923 or online at http://www.cganet.com.

6.1.2 MIL-SPECs. Military specifications are available at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.

6.1.3 MIL-STDs. Military standards are available at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.

6.2 National stock numbers (NSNs). The following is the NSN that is assigned to this CID. It may not be indicative of all possible NSNs associated with the CID.

4310-01-465-1509

6.3 Ordering data. The contract or order should specify the following:

a. CID document number and revision.
b. Product conformance provisions.
c. Packaging requirements.

6.4 Keywords.

Air charging system
SCBA

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MILITARY INTERESTS

Custodians:
Army – AT
Navy – SH
Air Force – 03

Review Activities:
Air Force – 84, 99
DLA – CC

Preparing Activity:
Navy – SH
(Project 4310-0044-000)

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